

Single Unit Auctions (private values)

- *First-price sealed-bid auction*: you don't see your opponents' bids. Highest bid wins. Winner pays her bid, b . The winner's profit is: $v - b$. Losers get nothing.
- *Second-price sealed-bid auction*: you don't see your opponents' bids. Highest bid wins. Winner pays *the second highest bid* in the auction. Therefore the winner's profit is: v minus the second highest bid. Losers get nothing.
- *Ascending-price drop-out auction*: Sort of like the game of chicken. Price starts ticking upwards from 0 until...wherever. You're all paying the price until you decide to drop out! As soon as the second-to-last person drops out, the auction ends. Winner's profit is v minus the price at which the second-to-last guy dropped out.

- *Descending price auction:* Game of chicken – in reverse! Price starts at \$11, ticks down until somebody says “I’ll take it.” The winner’s profit is v minus the price at which she called “I take it.”

Theoretical setup:

- N bidders draw valuations, v_i , from a joint distribution $F(v_1, \dots, v_N)$
- Simple setting: valuations are drawn independently from the same distribution $F(v)$.
- This is called the *symmetric independent private values* setting.
- Let's take $F(v) = v$, i.e. the uniform distribution on $[0, 1]$.

Second price auction:

- Claim: Regardless of what my competitors do, I can do no better than bidding my valuation.
- Reason 1: My payment does not depend on what I do.
- My payoff in every “contingency” is either
 - I win, and make $v_i - b^{2:N}$, where $b^{2:N}$ is the second highest bid.
 - I lose, and make zero.
 - My bid only affects whether I win or lose.
- If I bid my value, I do not regret winning or losing in any contingency.
- If I bid below my value, then I don’t win in some situations where I would want to win.

- If I bid above my value, I also win in some situations where I do NOT want to win.
- What is the expected revenue in 2nd price auction with N bidders, with uniform distributed valuations?
- Answer: $\frac{N-1}{N+1}$. Reasoning: N bidders with uniformly distributed valuations partition the interval of valuations to $N + 1$ equal segments; in expectation, of course (exercise: derive this).

Ascending auction:

- Claim: Regardless of what my competitors do, I can do no better than dropping out at my valuation.
- Reason: My payment does not depend on what I do.
- If I drop out below my value, then I don't win in some situations where I would be willing to win.
- If I drop out above my value, well, I make a loss.
- What is the expected revenue in ascending price auction with N bidders, with uniform distributed valuations?

First price auction:

- Won't bid valuation. Will want to “shade” your bid.
- How to optimally shade?

$$\max_b (v_i - b) \Pr\{\text{win}|b\}$$

How to calculate $\Pr\{\text{win}|b\}$?

Nash equilibrium:

- Suppose everybody plays the “strategy” $b(v)$, where $b(v)$ is an increasing function in v .
- Let’s look at the strategy, b_i , of bidder i , who has valuation v_i , assuming that all other bidders are using the strategy $b(v)$.

$$\max_{b_i} (v_i - b_i) \Pr\{b(v_j) \leq b_i, \forall j \neq i\}$$

$$\max_{b_i} (v_i - b_i) F^{N-1}(b^{-1}(b_i))$$

Take derivative w.r.t. b_i and set to zero

$$F^{N-1}(b^{-1}(b_i)) =$$

$$(v_i - b_i)(N - 1)F^{N-2}(b^{-1}(b_i))f(b^{-1}(b_i))\frac{1}{b'(b^{-1}(b_i))}$$

Now, v_i is not necessarily equal to $b^{-1}(b_i)$, but now, let's see what the above optimality condition tells us about the shape of $b(v)$ if bidder i also plays $b(v)$ (i.e. we're looking at a symmetric equilibrium of the game). In this case $b^{-1}(b_i) = v_i$, so

$$v_i - b(v_i) = \frac{1}{N - 1} \frac{F(v_i)b'(v_i)}{f(v_i)}$$

This is a first-order linear differential equation, with initial condition $b(0) = 0$. Solving this, we get:

$$b(v_i) = v_i - \frac{\int_0^{v_i} F^{N-1}(\theta)d\theta}{F^{N-1}(v_i)}$$

With uniform distribution, $F(v) = v$. So:

$$b(v) = \frac{N-1}{N}v$$

Expected revenue:

$$E[\text{Revenue}] = \frac{N-1}{N}E[\max\{v_i\}]$$

Remembering the “equal partitioning of the interval” trick:

$$E[\text{Revenue}] = \left(\frac{N-1}{N}\right)\left(\frac{N}{N+1}\right)$$

Observe that this is equal to the expected revenue from the second price auction! This was first observed by William Vickrey in 1961, and eventually got him a Nobel prize.

Descending price auction:

Claim: This is entirely equivalent to the first-price auction.

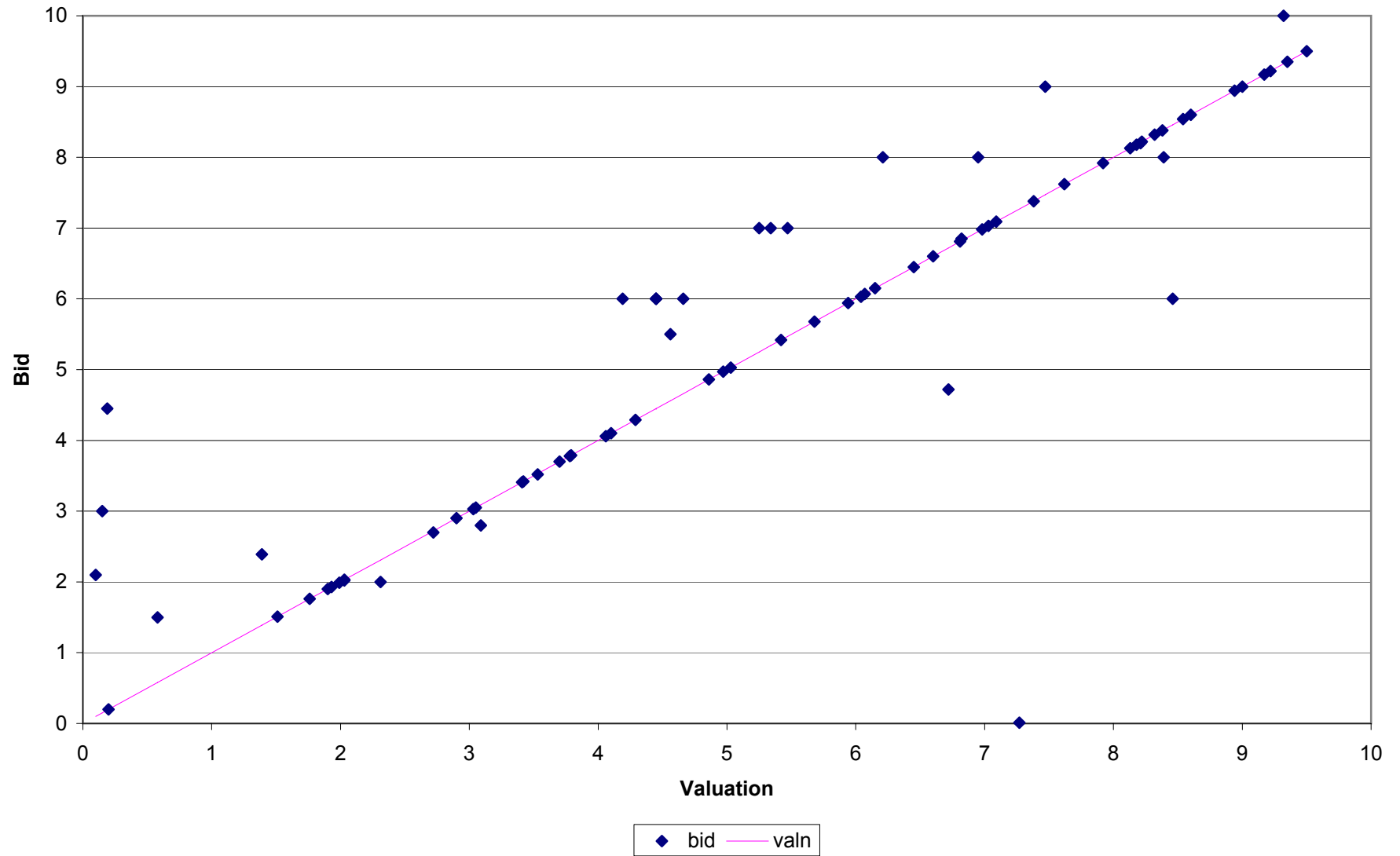
Why?

- Think of your strategy as pre-determining a price level at which you'll take the good.
- Will you update this pre-determined price-level during the auction? No, the only thing you see is that nobody has taken the good until now, but even if somebody takes it, it's too late!
- Call this strategy $b(v)$. But then the optimization problem solved by $b(v)$ is exactly the same as we did in the first-price auction.

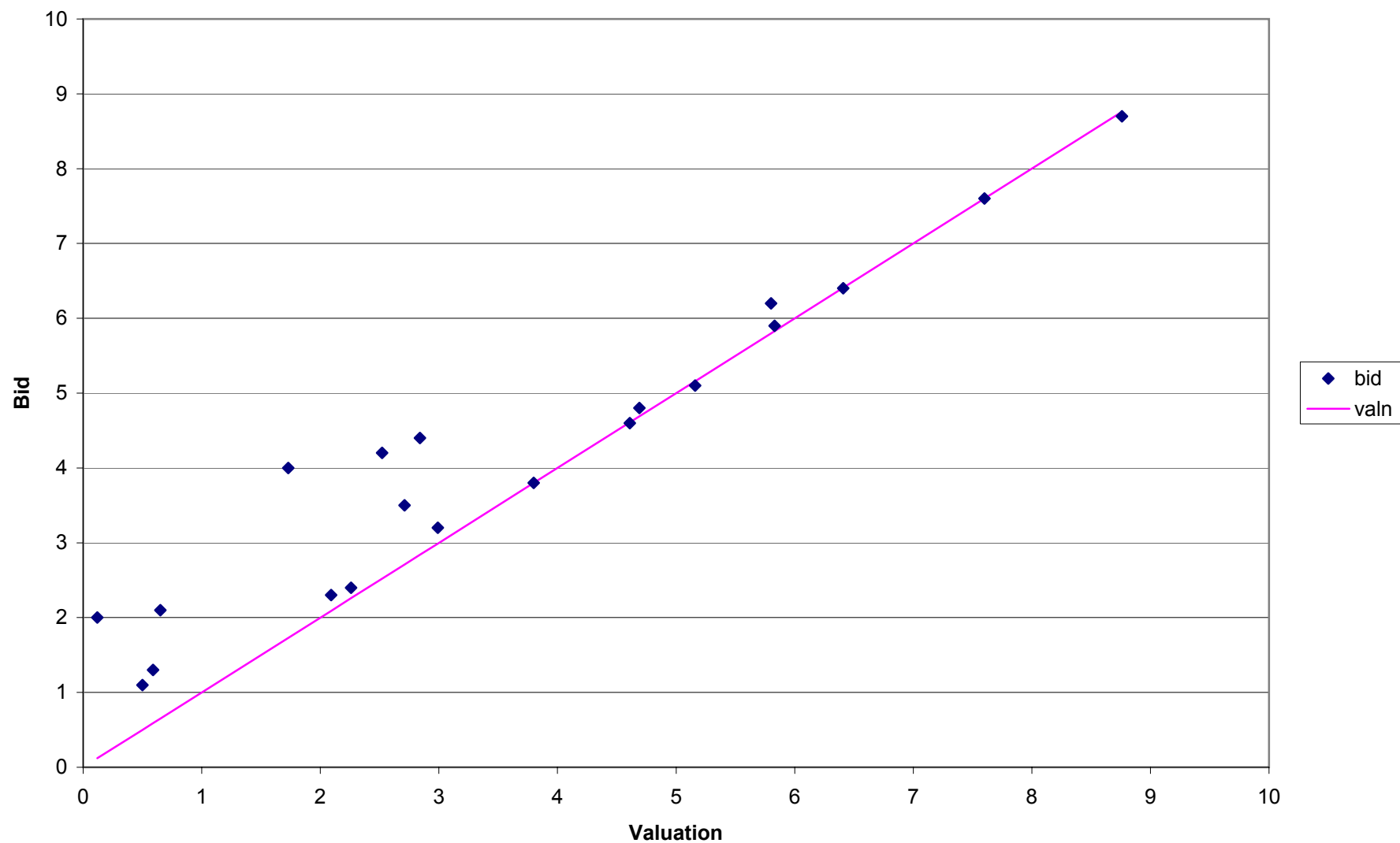
Summary:

- Although we have 4 different auctions, theory says they all yield the same expected revenue to the auctioneer.
- Also, theory says you should bid in the same way in the second-price and ascending auction
- And the same way in the first-price and descending auction

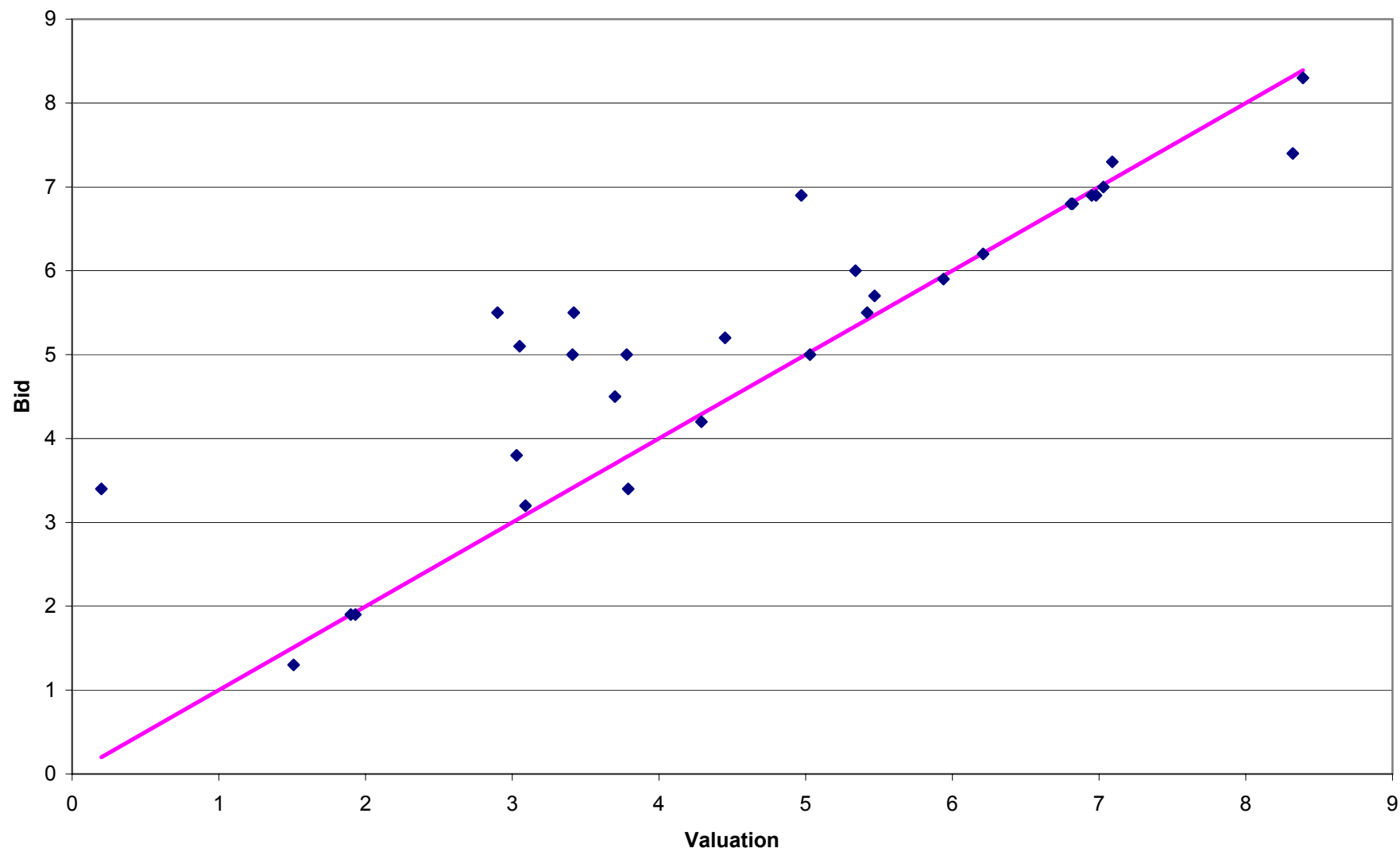
2nd price auction -- 2 bidders



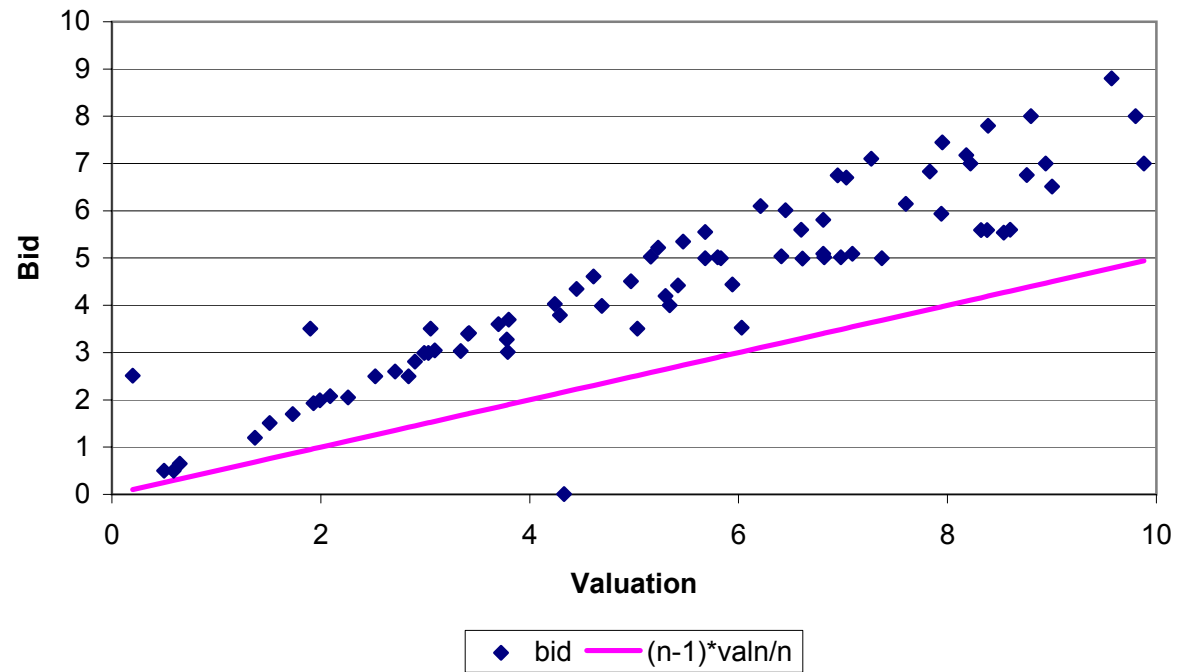
Ascending Price -- 2 bidders



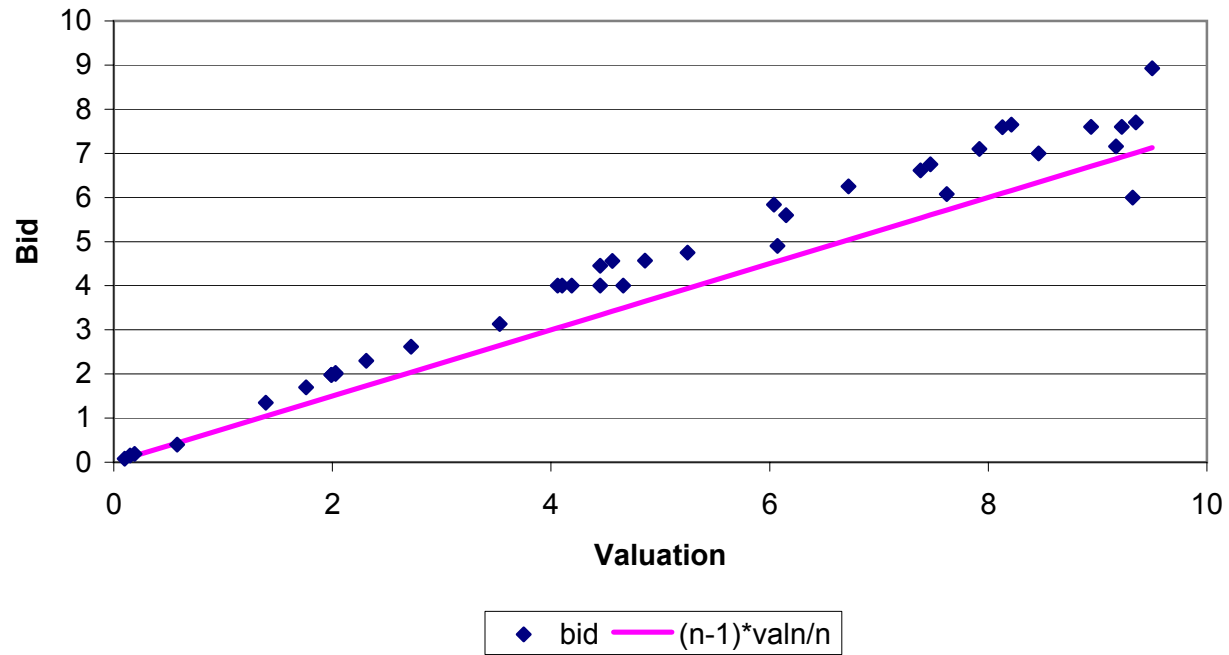
Ascending Price -- 4 bidders



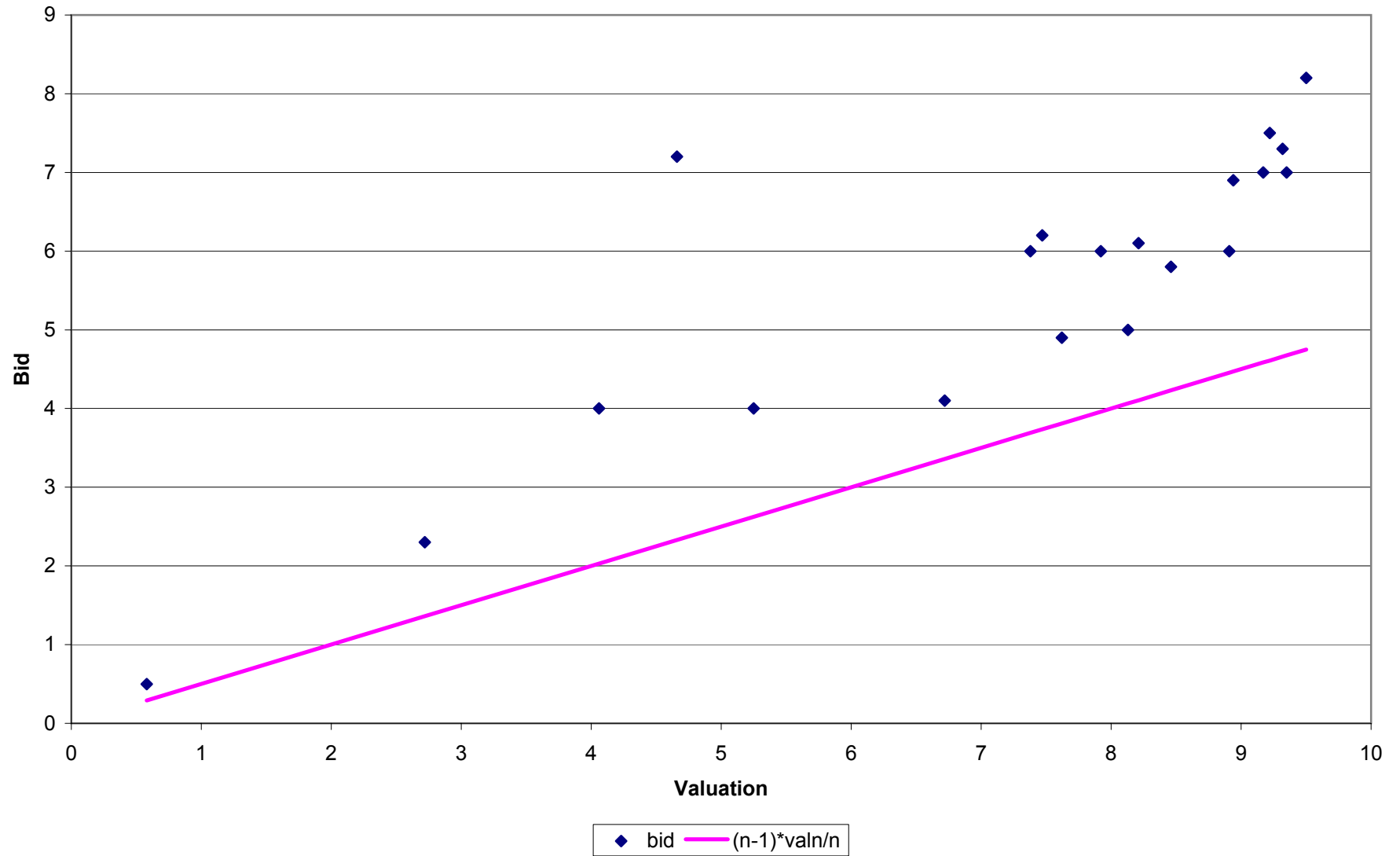
First price auction -- 2 bidders



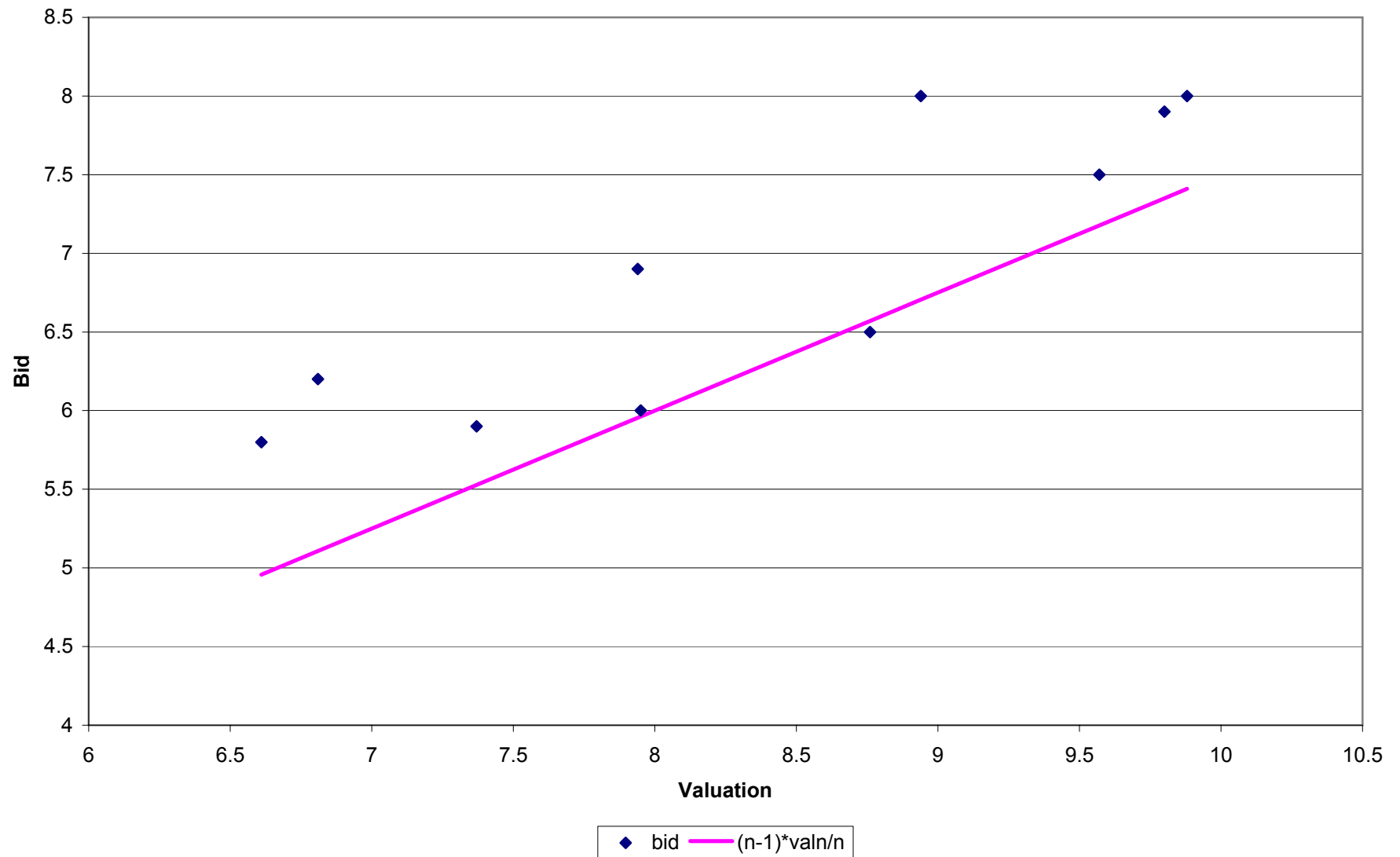
First price auction -- 4 bidders



Descending price -- 2 bidders



Descending Price Auction -- 4 bidders



auctiontype	mean revenue (N=2)	mean revenue (N=4)
1st price	5.53	7.05
2nd price	3.88	N/A
descending price	5.6	6.87
ascending price	4.18	6.47

Experimental evidence (rough guide):

- Subjects *learn* to bid their valuation in the ascending auction very quickly. Nobody wants to pay more than their valuation.
- Some people bid above their valuation in the second price auction. But, once they get hit by a bid coming between their valuation and their bid (and causing negative profit), they learn.
- I.e. people learn from their mistakes.
- The nice thing about the ascending auction is the “no-regret” feature. Ex-post nobody regrets having won the auction, or not winning it. This makes it a good auction procedure in many contexts. Plus, it’s really easy to strategize.
- Evidence from first-price auctions is a bit more mixed.

- Almost everybody finds an “overbidding” phenomenon.
- This is usually ascribed to risk aversion. If you lose, you get nothing. If you win, you get a strictly positive profit. So, if you’re risk-averse you want to take the “sting” out of losing by increasing your probability of winning by bidding a bit higher.
- Whatever the cause, it is clear that the first-price auction is a more complicated game than the ascending auction.
- Some previous experiments found that people bid a bit lower in the descending-price auction as opposed to first-price auction.
- I did not find this in a (small) experiment I ran. The differences in the literature are also not very large.
- My sense is that the same “risk-aversion” story applies here – people are afraid of losing, and hence will “jump” before the optimal point we derived under risk-neutrality.

- So, it looks like the first-price auction is a good way to extract revenue from risk-averse people.
- But, it may also cause some people to bid based on their risk aversion, as opposed to their valuation – causing inefficiencies in allocation.
- Disclaimer: A lot of what I said above is loose intuition – need to work through complicated models to say things rigorously.

Revenue Equivalence Theorem:

Proposition (Krishna 3.1): Suppose that values are distributed iid and bidders are risk neutral. Then any symmetric and increasing equilibrium of any standard auction, such that the expected payment of a bidder with value of zero is zero, yields the same expected revenue to the seller.

Proof:

- Fix auction format A , with equilibrium strategy $\beta(x)$ for value draw x .
- Let $m^A(x)$ be the equilibrium expected payment in auction A .
- Fix bidder 1 with value x and assume others are following β .
- Suppose bidder 1 bids $\beta(z)$ instead of $\beta(x)$. His expected

payoff is:

$$\Pi^A(z, x) = G(z)x - m^A(z)$$

- where $G(z) = F^{N-1}(z)$.
- Maximize wrt z :

$$\frac{\partial}{\partial z} \Pi^A(z, x) = g(z)x - \frac{d}{dz} m^A(z) = 0$$

- At equilibrium it is optimal to set $z = x$, so:

$$\frac{d}{dz} m^A(y) = g(y)y$$

- So, by integrating, we get:

$$\begin{aligned} m^A(x) &= \int_0^x yg(y)dy \\ &= \int_0^x y(N-1)F^{N-2}(y)f(y)dy \end{aligned}$$

- Notice that this only depends on $F(\cdot)$ and N , and not on β .

It turns out this can be much more general.

- In any (private value) auction, a bidder's choice of bid affects only his expected payment, or his probability of winning.
- That is, by choosing your bid, you're choosing π , the probability you win the auction.
- The auction rules specify $e(\pi)$, the expected payment you make.
- Your utility is (under risk-neutrality) $u = v\pi - e(\pi)$
- “Strategic behavior” amounts to differentiation w.r.t. π
- This yields: $e'(\pi) = v$
- But π is also a function of v . In all the auctions above, $\pi(v)$ is an increasing function of v ; the person with the higher valuation has higher probability of winning.
- In fact, in all the auction above, $\pi(v)$ is the same! The person

with the highest valuation wins in all auctions, so $\pi(v)$ is just the probability that you have the highest valuation!

- But if $e'(\pi(v)) = v$ (due to strategic behavior), we can write:

$$\begin{aligned} e(\pi(x)) &= e(\pi(0)) + \int_{\pi(0)}^{\pi(x)} \frac{de(\pi')}{d\pi'} d\pi' \\ &= e(\pi(0)) + \int_0^x \frac{de(\pi(\theta))}{d\pi} \frac{d\pi(\theta)}{d\theta} \\ &= e(p(0)) + \int_0^x \theta \frac{dp(\theta)}{d\theta} \end{aligned}$$

But we just said above that $\pi(\theta)$ is the same for all auctions. I.e. the expected payment of a bidder depends ONLY on the probability that he wins the auction with a given valuation.

Corollary: All (independent private value) auctions that award the

good to the highest bidder, and pay zero to bidder with lowest valuation yield the same expected payments to all bidders, and hence yield the same expected revenue for the auctioneer.

This is the Revenue Equivalence Theorem.

Bottomline: Under risk-neutrality, if bidders' valuations are independent, two auction rules that lead to the same probability of winning to a bidder with valuation v (for all valuations v) lead to the same expected payments for all bidders, and hence lead to the same expected revenue for the auctioneer.

An application: the “all-pay” auction:

- This is a widely used a model of bribing, lobbying, campaigning, and patent races.
- Highest bidder wins, but everybody pays their bid.
- Is $p(v)$ different in this context?
- In particular, is $p(v)$ different than in the second price auction?
What does this imply about $e(p(v))$ across the two auctions?
- What was $e(p(v))$ in second price auction, with uniformly

distributed values (expected payment of a bidder with valuation v)?

- Claim: In second price auction $e(p(v)) = \frac{N-1}{N}v \times v^{N-1}$
- I.e. expected payment conditional on winning, times probability of winning.
- In the all-pay auction, your expected payment is equal to your bid.
- Hence $b(v) = \frac{N-1}{N}v^N$ is the bidding strategy in the all-pay auction.

Risk aversion

Claim: expected revenue under first-price auction is higher than in second price auction.

- No change in equilibrium strategies in second price auction.
- Assume symmetric increasing equilibrium $\gamma(x)$ in first price auction:

$$\max_z F^{N-1}(z)u(x - \gamma(z))$$

- First order condition wrt “reported valuation”

$$g(z)u(x - \gamma(z)) - G(z)\gamma'(z)u'(x - \gamma(z)) = 0$$

- In symmetric equilibrium, it's optimal to set $z = x$, so:

$$\gamma'(x) = \frac{u(x - \gamma(x))}{u'(x - \gamma(x))} \frac{g(x)}{G(x)}$$

- with risk neutrality:

$$\beta'(x) = (x - \beta(x)) \frac{g(x)}{G(x)}$$

- If $u(0) = 0$, and $u(\cdot)$ strictly concave, $u(y) > u'(y)y$, so:

$$\frac{u(x - \gamma(x))}{u'(x - \gamma(x))} \frac{g(x)}{G(x)} > x - \gamma(x) \frac{g(x)}{G(x)}$$

- So $\gamma(x) < \beta(x)$ implies $\gamma'(x) > \beta'(x)$
- $\gamma(0) = \beta(0) = 0$, so the above is a contradiction, i.e.

$$\gamma(x) > \beta(x)$$

- Risk aversion makes people bid higher.

Asymmetries (Maskin and Riley, 2000)

Bidders are *ex-ante* asymmetric.

- Two bidders, F_1 on $[0, \omega_1]$ and F_2 on $[0, \omega_2]$.
- Assume increasing bid strategies, β_1 and β_2 , with inverses, $\phi_1 \equiv \beta_1^{-1}$ and $\phi_2 \equiv \beta_2^{-1}$.
- $\beta_1(0) = 0 = \beta_2(0)$.
- $\beta_1(\omega_1) = \beta_2(\omega_2)$ (Why?)
- Bidder i 's maximization problem

$$\max_b \Pi_i(b, x_i) = F_j(\phi_j(b))(x_i - b)$$

- FOC wrt b

$$\phi'_j(b) = \frac{F_j(\phi_j(b))}{f_j(\phi_j(b))} \frac{1}{\phi_i(b) - b}$$

- I.e. coupled differential equations for $\phi_1(b)$ and $\phi_2(b)$.
- Hard to solve analytically in general.
- Special case: $\omega_1 \geq \omega_2$ and

$$\frac{f_1 x}{F_1(x)} > \frac{f_2 x}{F_2(x)}$$

- this also implies stochastic dominance (not the other way around)
- Proposition (Maskin and Riley): In this case $\beta_2(x) > \beta_1 x$. I.e. for a given valuation, the weak bidder bids more aggressively than the strong bidder!
- This means that the auction outcome need no longer be

efficient.

- Indeed, one can construct examples where the second price auction revenue dominates the first price auction, and vice versa. I.e. revenue equivalence no longer holds!

Proof of Maskin-Riley proposition

- Suppose $\beta_2(z) = \beta_1(z) = c$ at some z .
- Then $\phi_2(c) = \phi_1(c) = z$. But then

$$\phi'_2(c) = \frac{F_2(z)}{f_2(z)} \frac{1}{z - b} > \frac{F_1(z)}{f_1(z)} \frac{1}{z - b} = \phi'_1(c)$$

- $\phi'_i(c) = 1/\beta'_i(z)$, so this means $\beta'_1(z) > \beta'_2(z)$.
- I.e. the curves intersect only once.
- Now suppose $\beta_1(x) \geq \beta_2(x)$ for some x . We'll construct contradictions to this.
- Then, either $\beta_1(x) > \beta_2(x)$ for all x , or they intersect once at some z , but $\beta_1(x) > \beta_2(x)$ for $x > z$.
- But if (by assumption) $\omega_1 > \omega_2$, $\beta_1(\omega_1) = \beta_2(\omega_2)$, so $\beta_1(\omega_2) < \beta_2(\omega_2)$. Contradiction.

- Look at the paper for the case of common upper support.

Econometrics of Single Unit Auctions

- (Sub)set of bids from a sequence of auctions
- Structural “parameter:” $F_V(v)$
- Cool thing about auctions: the “error” terms are given explicitly by model.
- Second-price auction: suppose you observe all bids
- $v = b$. Done. $G_b(b) = F_V(b)$.
- Suppose you only observe winner’s payment.
- Assume symmetry: $G_{b^{2:N}}(x) = (N - 1)F_V^{N-1}(x)(1 - F_V(x))$
- Solve for $F_V(x)$
- In general (Haile and Tamer (2003), Athey and Haile (2002)):
- Given iid random variables drawn from $F(v)$, the distribution

of the i th order statistic, $F_{i:N}(v)$ is related to the parent distribution by (the monotonic function):

$$F_{i:N}(v) = \frac{N!}{(N-i)!(i-1)!} \int_0^{F(v)} s^{i-1} (1-s)^{N-i} ds$$

Haile and Tamer (JPE 2003)

“Inference with an Incomplete Model of English Auctions”

- Data from ascending auctions.
- But don't want to believe that drop out points of losing bidders necessarily reflect their valuations.
- “Incomplete” behavioral assumptions:
- A1: Bidders do not bid more than they are willing to pay.
- A2: Bidders do not allow an opponent to win at a price they are willing to beat.
- A1 is equivalent to: $b_i \leq v_i \forall i$
- OR: $G(v) \geq F(v)$ (bid distribution is stochastically dominated by the value distribution).
- The same stochastic dominance applies to all order statistics.

- I.e. a whole family of *upper* bounds for $F(v)$.
- A2 is equivalent to: $v_{N-1:N} \leq b_{N:N} + \Delta$
- with the dominance relation: $F_{N-1:N}(v) \geq G_{N:N}(v)$,
- which, through the inversion formula, gives a *lower* bound on $F(v)$.

Estimation

Use empirical distribution functions:

$$\hat{G}_{i:n}(v) = \frac{1}{T_N} \sum_{t=1}^T 1\{N_t = N, b_{i:N_t} \leq v\}$$

and invert using the $\phi(\cdot)$ mapping. Uniform consistency obtains using the Glivenko-Cantelli and Slutsky theorems.

When there is heterogeneity across auctions:

- $E[V_i t | X_t = x_t] = l(x_t, \beta_0)$ where $l(\cdot, \cdot)$ is known.
- Then $E[\underline{S}_x] \leq l(x, \beta_0) \leq E[\overline{S}_x]$ where $\underline{S}_x$ is distributed with upper bound cdf, and $\overline{S}_x$ is distributed with lower bound cdf.
- The set of parameter values b indistinguishable from β_0 is given by:

$$\{b : \Pr[\{x : l(x, b) < E[\underline{S}_x] \cup E[\overline{S}_x] < l(x, b)\}] = 0\}$$

Set estimation criterion function:

$$\begin{aligned} Q(b) = & \int [E[\underline{S}_x] - l(x, b)]^2 \delta_1(x, b) \\ & + [l(x, b) - E[\overline{S}_x]]^2 \delta_2(x, b) dP_X \end{aligned}$$

where

$$\delta_1(x, b) = 1\{E[\underline{S}_x] > l(x, b)\}$$

$$\delta_2(x, b) = 1\{E[\overline{S}_x] < l(x, b)\}$$

Minimize this to get b . (Not quite, actually, so read the paper.)

Econometrics of First Price Auctions

- Seems not as easy since $b(v)$ is a nonlinear function.
- One solution: parameterize $F_V(v, \theta)$.
- For each bid b_i , find the value, $v_i = b^{-1}(b_i; \theta)$, that rationalizes this bid for a given value of θ . Thus, the likelihood of observing b_i is $f_V(b^{-1}(b_i; \theta))$, where the mapping $b(v)$ is the solution of the ODE:

$$b(v_i) = v_i - \frac{\int_0^{v_i} F^{N-1}(\theta) d\theta}{F^{N-1}(v_i)}$$

One actually write the ODE in terms of the inverse bid function $\phi(b(v)) = b^{-1}(b(v))$, which makes life easier.

$$\phi'(b) = \frac{1}{N-1} \frac{F(\phi(b))}{f(\phi(b))} \frac{1}{\phi(b) - b}$$

- Estimate using maximum likelihood.
- How to do this? Given θ , solve for $\phi(b|\theta)$.
- Using ϕ , associate each bid with a valuation.
- Likelihood of observed bids = likelihood of valuations (times Jacobian)
- Disadvantage 1: Have to solve the ODE for every θ iteration.
- Disadvantage 2: How do we know whether parametric assumption is good?
- Disadvantage 3: Support of the distribution of values depends on support of bids. The convergence of estimator no longer $\sqrt{N}$ (actually faster), thus can't use MLE standard error

formula. Need parametric bootstrap to calculate standard errors. OR, use in a Bayesian MCMC estimation framework.

Guerre, Perrigne and Vuong (2000, EMA)

$$\max_b (v_i - b) \Pr\{\text{win}|b\}$$

$$(v_i - b) = \frac{\Pr\{\text{win}|b\}}{\frac{\partial \Pr\{\text{win}|b\}}{\partial b}}$$

In the first-price auction, the FOC becomes:

$$v_i - b = \frac{1}{N-1} \frac{G(b)}{g(b)}$$

(First step) Estimate v_i by:

$$\hat{v}_i - b = \frac{1}{N-1} \frac{\hat{G}(b)}{\hat{g}(b)}$$

and (second step) estimate $F_V(v)$ by the empirical cdf of $\hat{v}$.

Estimators (for data with L auctions):

$$\tilde{G}(b) = \frac{1}{NL} \sum_{l=1}^L \sum_{p=1}^N 1\{B_{pl} \leq b\}$$

$$\tilde{g}(b) = \frac{1}{NLh_g} \sum_{l=1}^L \sum_{p=1}^N K_g\left(\frac{b - B_{pl}}{h_g}\right)$$

where h_g is a bandwidth parameter and $K_g(.)$ is a kernel with compact support. Issue: kernel estimator of density is biased at the boundaries of the support.

Cross-Auction Heterogeneity

- Data: bids from $l = 1, \dots, L$ auctions, with N_l bidders per auction. Additional auction level covariate vector X_l , which has d dimensions.

$$\begin{aligned} \tilde{G}(b, x, n) = \\ \frac{1}{L h_G^d} \sum_{l=1}^L \frac{1}{N_l} \sum_{p=1}^{N_l} 1\{B_{pl} \leq b\} K_G\left(\frac{x - X_l}{h_G}, \frac{n - N_l}{h_{GN}}\right) \end{aligned}$$

$$\begin{aligned} \tilde{g}(b, x, n) = \\ \frac{1}{L h_g^{d+1}} \sum_{l=1}^L \frac{1}{N_l} \sum_{p=1}^{N_l} K_g\left(\frac{b - B_{pl}}{h_g}, \frac{x - X_l}{h_G}, \frac{n - N_l}{h_{GN}}\right) \end{aligned}$$

Problem: curse of dimensionality. You can't let dimension of covariate vector grow very large.

Athey, Levin and Seira (2004)

They look at timber auctions, where:

- Bidders are heterogeneous
- Both open and sealed formats adopted

Two formats of auction: Open ascending and Sealed bid

⇒ What will be the difference of two formats?

- participation decisions
 - allocation
 - revenue
-
- Can model explain the difference in data?
 - How can we recover bidders' valuations?
 - How can we use such valuations to quantify the difference?

Athey, Levin and Seira (2004) cont'd

- Format: IPV, sealed or open ascending
- Bidders: heterogeneous, risk-neutral
- Data: U.S. Forest Service Timber Auctions

U.S. Forest Service Timber Program

- Use both open ascending and sealed auctions
- Timber logging and milling: \$100 billion a year industry
- U.S. Federal government sold a billion dollars of timber a year.

Timber Auctions: Timeline

- Forest Service identifies a tract, organizes a “cruise”
- Announce
 - estimates of available timber, logging costs
 - tract characteristics
 - reserve price
 - open or sealed bid
- Bidders decide whether to pay cost K to learn valuation.
- Participating bidders know who gathered information.
- Each bidder who acquired information submit bids

Theory

- Baseline: Maskin and Riley (2000)
- Further,
 - (i) endogenize participation
 - (ii) allow collusion in open auction

Two types of bidders

- “mills”

bidders who have manufacturing capacity

- “loggers”

bidders who do not.

Notations

- N potential bidders
- Each bidder i has private value v_i
- Loggers have value distribution: F_L
- Mills' value distribution: F_M
- A strategy for i
 - Bidding strategy: $b_i(\cdot; n)$
 - Entry strategy: enter iff entry cost (k_i) is below his entry cost threshold (K_L or K_M)
- n participants

Assumption and Equilibrium

Assumption (i) F_L, F_M have continuous densities f_L, f_M ; and (ii) for all v , $\frac{f_M(v)}{F_M(v)} \geq \frac{f_L(v)}{F_L(v)}$.

A *type-symmetric entry equilibrium* is a pair of bidding strategies $b_L(\cdot; n), b_M(\cdot; n)$ and entry cost thresholds K_L, K_M with the property that: (i) loggers use the strategy b_L, K_L and mills the strategy b_M, K_M ; (ii) each bidder's bid strategy maximizes his profits conditional on entering; and (iii) each bidder finds it optimal to enter if and only if his entry cost lies below his cost threshold.

Sealed bid auction

Suppose i is a participating bidder with value v_i . His expected profit is

$$\pi_i^s(v_i; n) := \max_{b \geq r} (v_i - b) \prod_{j \in n \setminus i} G_j(b; n), \quad (1)$$

where $G_j(b; n)$ is the probability that j will bid less than b . In equilibrium, bid strategies will be continuous and strictly increasing, so $G_j(b; n) = F_j(b_j^{-1}(b; n))$.

The first order condition for i 's bidding problem is

$$\frac{1}{v_i - b_i} = \sum_{j \in n \setminus i} \frac{g_j(b_i; n)}{G_j(b_i; n)}. \quad (2)$$

Given a set of entry thresholds K , bidder i 's expected profit from entry is

$$\Pi_i^s(K) = \sum_{n \subset N} \left\{ \int \pi_i^s(v_i; n) dF_i(v_i) \right\} \Pr[n \mid K, i \in n, s],$$

In equilibrium, entry thresholds are equal to expected profit for each bidder

Open Auctions

- It is a dominant strategy to bid his value
- The bidder i 's expected profit conditional on entering:

$$\pi_i^o(v_i; n) : \max_{b \geq r} (v_i - \mathbb{E}[\max\{v_{-i}, r\} | v_j \leq b \ \forall j \in n \setminus i]) \prod_{j \in n \setminus i} F_j(b). \quad (4)$$

Bidder i 's expected profit as a function of the entry cost thresholds is

$$\Pi_i^o(K) = \sum_{n \subset N} \left\{ \int \pi_i^o(v_i; n) dF_i(v_i) \right\} \Pr[n \mid K, i \in n, o].$$

In equilibrium, $K_i^o = \Pi_i^o(K^o)$.

Theoretical Result 1: Sealed Bid Auctions

Proposition 1 *A type-symmetric entry equilibrium exists in the sealed bid auction. In equilibrium: (i) mills submit higher bids: $G_M(b; n) \leq G_L(b; n)$ for all b , despite the fact that (ii) mills shade their bids more than loggers: $b_M(v; n) \leq b_L(v; n)$ for all v .*

Theoretical Result 2: Ascending Auctions

Proposition 2 *A type-symmetric entry equilibrium exists in the open bid auction. In equilibrium, (i) mills submit higher bids; and (ii) all entrants bid their true value: $b_i(v; n) = v$ for all v .*

Theoretical Result 3: Revenue Equivalence

Proposition 3 (*Revenue Equivalence*) *If bidders are homogenous, so $F_L = F_M$, the sealed bid and open auction each have a unique symmetric entry equilibrium, in which the highest valued entrant wins the auction. These equilibria have (i) the same expected entry, and (ii) the same expected revenue.*

Theoretical Result 4 : Comparing Formats

Proposition 4 *For any type-symmetric entry equilibrium of the sealed bid auction, there is a type-symmetric entry equilibrium of the open auction in which: (i) loggers are less likely to enter; (ii) mills are more likely to enter; (iii) it is less likely a logger will win.*

Theoretical Result 5 : Efficiency

Proposition 5 (*Efficiency*) *The socially efficient type-symmetric strategy profile is an entry equilibrium of the open auction, but every sealed auction equilibrium is inefficient.*

No Theoretical Prediction : Revenue

Theoretical Result 5 : Collusion

Proposition 6 *(Collusion)* For any type-symmetric entry equilibrium of the open auction, there is a type-symmetric collusive equilibrium in which: (i) Loggers are less likely to enter; (ii) Mills are more likely to enter; (iii) It is less likely a logger will win. Thus, for any type-symmetric entry equilibrium of the sealed bid auction, there is a type-symmetric collusive equilibrium of the open auction where (i)-(iii) hold.

Findings

- Sealed bidding induces significantly more participation by loggers
- Sealed auctions are more likely to won by loggers
- Winning bids
 - 12-18% higher in the sealed bid in Northern forests
 - small and not significant difference in California forests

Data

- Regions:
 - Northern region comprised of Idaho and Montana
 - California
- Sample periods: 1982-1990

Table 1A: Summary Statistics for Northern Sales

N	Open Auctions				Sealed Auctions			
	Full Sample 787		Selected 658		Full Sample 308		Selected 300	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
<i>Auction Outcomes</i>								
Winning Bid (\$/mbf)	70.14	52.94	72.78	53.81	80.21	56.25	81.10	56.57
Entrants	4.12	2.46	4.23	2.45	4.53	2.84	4.57	2.86
# Loggers Entering	2.62	2.40	2.84	2.39	3.36	2.58	3.42	2.59
# Mills Entering	1.50	1.65	1.40	1.66	1.17	1.66	1.14	1.66
Fraction Loggers Entering	0.61	0.39	0.65	0.38	0.76	0.32	0.77	0.32
Logger Wins Auction	0.56	0.50	0.60	0.49	0.67	0.47	0.69	0.46
<i>Appraisal Variables</i>								
Volume of timber (hundred mbf)	28.93	39.64	21.95	33.71	15.02	26.97	12.88	22.51
Reserve Price (\$/mbf)	26.22	26.72	27.45	27.72	28.46	24.24	28.68	24.38
Selling Value (\$/mbf)	196.04	168.41	196.02	169.11	202.59	166.07	201.80	166.66
Road Construction (\$/mbf)	6.36	9.84	4.91	9.07	3.11	7.77	2.83	7.54
No Road Construction	0.58	0.49	0.66	0.47	0.78	0.42	0.79	0.41
Logging Costs (\$/mbf)	84.66	63.64	82.91	63.77	83.55	62.81	82.51	63.25
Manufacturing Costs (\$/mbf)	114.59	84.04	112.93	84.71	117.79	85.57	116.75	86.40
<i>Sale Characteristics</i>								
Contract Length (months)	24.78	17.38	22.19	16.35	18.12	14.79	17.03	13.11
Species Herfindal	0.60	0.27	0.59	0.28	0.58	0.27	0.58	0.27
Density of Timber (hmbf/acres)	0.07	0.06	0.07	0.06	0.08	0.07	0.08	0.07
Salvage Sale	0.37	0.48	0.37	0.48	0.39	0.49	0.40	0.49
Scale Sale	0.44	0.50	0.42	0.49	0.41	0.49	0.40	0.49
Quarter of Sale	2.39	1.00	2.39	1.01	2.42	1.01	2.42	1.01
Year of Sale	86.08	2.31	86.07	2.38	85.75	2.52	85.76	2.55
Housing Starts	1580.62	237.95	1572.33	235.52	1559.18	261.09	1553.84	261.71
<i>Potential Competition</i>								
Logging companies in county	43.86	21.22	42.15	21.67	40.05	22.22	40.36	22.35
Sawmills in County	8.66	4.45	8.42	4.56	7.60	4.47	7.45	4.30
Active Loggers (active in District in prior 12 months)	30.97	24.83	30.19	24.22	25.83	17.62	26.19	17.69
Active Manufacturers (active in District in prior 12 months)	11.02	9.01	11.50	9.26	12.33	10.30	12.54	10.34

Table 1B: Summary Statistics for California Sales

N	Open Auctions				Sealed Auctions			
	Full Sample 1188		Selected 212		Full Sample 694		Selected 269	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Auction Outcomes								
Winning Bid (\$/mbf)	108.62	165.23	118.95	103.51	93.25	71.80	92.09	74.24
Entrants	4.13	2.32	4.23	2.41	3.85	2.59	4.40	2.68
# Loggers Entering	1.15	1.56	2.12	2.09	2.86	2.25	3.02	2.35
# Mills Entering	2.98	1.81	2.11	1.90	0.99	1.43	1.38	1.58
Fraction Loggers Entering	0.24	0.28	0.50	0.37	0.77	0.31	0.70	0.32
Logger Wins Auction	0.17	0.38	0.43	0.50	0.73	0.45	0.62	0.49
Appraisal Variables								
Volume of timber (hundred mbf)	63.63	45.60	19.85	20.00	7.39	13.38	10.46	10.36
Reserve Price (\$/mbf)	41.96	38.02	49.68	46.54	42.56	39.84	37.32	37.09
Selling Value (\$/mbf)	278.86	85.30	246.80	131.93	234.49	268.00	247.68	118.60
Road Construction (\$/mbf)	10.66	12.95	4.71	11.44	1.08	4.33	2.04	5.89
No Road Construction	0.26	0.44	0.67	0.47	0.90	0.29	0.83	0.38
Logging Costs (\$/mbf)	112.85	40.48	96.24	55.24	89.15	56.32	103.47	52.70
Manufacturing Costs (\$/mbf)	127.41	34.47	109.20	54.36	100.97	61.85	114.06	52.95
Sale Characteristics								
Contract Length (months)	28.68	14.35	16.37	9.75	10.01	6.62	12.51	6.16
Species Herfindal	0.54	0.23	0.59	0.25	0.60	0.24	0.58	0.24
Density of Timber (hmbf/acres)	0.10	0.12	0.09	0.10	0.16	1.82	0.11	0.15
Salvage Sale	0.14	0.35	0.25	0.43	0.36	0.48	0.26	0.44
Scale Sale	0.95	0.21	0.86	0.35	0.67	0.47	0.82	0.38
Quarter of Sale	2.35	1.00	2.55	0.95	2.71	0.88	2.65	0.93
Year of Sale	85.32	2.14	85.62	2.42	85.59	2.30	85.01	2.15
Housing Starts	1587.06	251.78	1528.56	260.22	1558.48	249.87	1581.44	264.07
Potential Competition								
Logging companies in county	23.22	18.65	22.32	17.56	20.39	17.35	23.06	19.84
Sawmills in County	6.65	6.50	6.14	5.55	6.05	6.01	7.04	7.73
Active Loggers (active in Forest in prior 12 months)	57.65	32.79	60.23	31.55	54.37	30.31	57.96	28.28
Active Manufacturers (active in Forest in prior 12 months)	47.39	27.81	48.96	26.17	44.48	27.08	46.48	26.25

Comparing Auctions: Evidence

Approach

$$\begin{aligned} Y &= f(SEALED, X, N, \varepsilon) \\ &= \alpha \cdot SEALED + X\beta + N\gamma + \varepsilon \end{aligned}$$

We want to know

$$\gamma_Y = E_{X, \varepsilon} [f(1, X, N, \varepsilon) - f(0, X, N, \varepsilon)]$$

Table 3A: Effect of Auction Method on Sale Outcomes (Northern Sales)

(N= 958 Sales)

Dependent Variable:	(1) ln(Logger Entry)	(2) ln(Mill Entry)	(3) Loggers/Entrants	(4) Logger Wins	(5) ln(Price)	(6) ln(Price) ¹
<i>Regression with No Interactions Between Sealed and Covariates²</i>						
Sealed Bid Effect	0.104 (0.037)**	-0.017 (0.032)	.056 (0.016)***	0.044 (0.028)	0.125 (0.039)***	0.076 (0.032)**
<i>Regression with Interactions Between Sealed and All Covariates</i>						
Sealed Bid Effect on Sample	0.105 (0.037)**	0.004 (0.033)	0.045 (0.015)**	0.034 (0.028)	0.139 (0.041)***	0.087 (0.032)*
<i>Matching Estimate³</i>						
Sealed Bid Effect on Sample	0.158 (0.043)***	-0.036 (0.041)	0.079 (0.019)***	0.074 (0.032)*	0.179 (0.052)**	0.133 (0.048)**

Notes: Robust standard errors in parentheses.

1. Specification includes number of entering mills and loggers in addition to sale controls.

2. See Appendix Tables 1A and 2A for full set of controls and coefficients.

3. Number of matches = 4 using same controls as regression estimates and the estimated propensity score.

Table 3B: Effect of Auction Method on Sale Outcomes (California Sales)

(N= 481 Sales)

<i>Dependent Variable:</i>	(1) ln(Logger Entry)	(2) ln(Mill Entry)	(3) Loggers/Entrants	(4) Logger Wins	(5) ln(Price)	(6) ln(Price) ¹
<i>Regression with No Interactions Between Sealed and Covariates²</i>						
Sealed Bid Effect	0.131 (0.058)*	-0.069 (0.051)	0.087 (0.029)**	0.086 (0.046)+	0.013 (0.065)	-0.048 (0.055)
<i>Includes Interactions Between Sealed and All Covariates</i>						
Sealed Bid Effect on Sample	0.120 (0.058)*	-0.079 (0.050)	0.084 (0.029)**	0.077 (0.046)+	0.008 (0.064)	-0.027 (0.048)
<i>Matching Estimate³</i>						
Sealed Bid Effect on Sample	0.181 (0.061)**	-0.194 (0.053)***	0.152 (0.031)***	0.135 (0.045)**	-0.048 (0.076)	-0.027 (0.075)

Notes: Robust standard errors in parentheses.

1. Specification includes number of entering mills and loggers in addition to sale controls.

2. See Appendix Tables 1B and 2B for full set of controls and coefficients.

3. Number of matches = 4 using same controls as regression estimates and the estimated propensity score.

Structural Estimation

- Estimate value distribution of mills and loggers (Guerre, Perrigne and Vuong (2000))
 - Fit a distribution to the observed sealed bid data, yield bid distribution $G_L(\cdot / X, u, N, n), G_M(\cdot / X, u, N, n)$
 - Use FOC to recover bidder's value distribution $F_L(\cdot / X, u, N), F_M(\cdot / X, u, N)$
- Introduce unobserved heterogeneity u
- Assume bidders have entry costs $K(X, N)$

Estimate Bid Distributions

- Estimation: Parametric specifications
 - Bid distributions: Weibull distribution
 - Tract heterogeneity, u : Gamma distribution

$k = L, M$:

$$G_k(b|X, u, N, n) = 1 - \exp \left(-u \cdot \left(\frac{b}{\lambda_k(X, N, n)} \right)^{p_k(n)} \right). \quad (8)$$

Here $\lambda_k(\cdot)$ is the scale, and $p_k(\cdot)$ the shape, of the Weibull distribution, parametrized as $\ln \lambda_k(X, N, n) = X\beta_X + N\beta_N + n\beta_{n,k} + \beta_{0,k}$ and $\ln p_k(n) = n\gamma_{n,k} + \gamma_{0,k}$.^{30,31}

Estimate Bid Distributions (cont'd)

- To include rich set of covariates and unobserved heterogeneity, choose parametric approach.
- Guerre, Perrigne and Vuong (2000): Non-parametric
- Krasnokutskaya (2002): Beta-Weibull

Table 4A: Bid and Entry Distributions for Sealed Auctions (Northern Sales)

	(1)		(2)	
	Bid Distribution (Weibull)		Logger Entry (Poisson)	
	coefficient	s.e.	coefficient	s.e.
	<i>ln(λ)</i>		<i>ln(μ)</i>	
Ln(Reserve Price)	0.42	(0.04)	-0.29	(0.05)
Ln(Selling Value)	-0.01	(0.02)	-0.03	(0.03)
Ln(Manufacturing Costs)	0.44	(0.14)	0.85	(0.17)
Ln(Logging Costs)	-0.44	(0.14)	-0.81	(0.17)
Ln(Road Costs)	0.00	(0.02)	-0.16	(0.04)
Species Herfindal	-0.10	(0.11)	-0.24	(0.15)
Density of Timber (hmbf/acres)	-0.96	(0.31)	-0.93	(0.44)
Salvage Sale (Dummy)	-0.05	(0.05)	-0.02	(0.07)
Scale Sale (Dummy)	-0.05	(0.05)	-0.15	(0.08)
Ln(Volume)	-0.08	(0.03)	-0.24	(0.04)
Kootenai NF (Dummy)	0.14	(0.06)	0.18	(0.09)
Mill (Dummy)	0.27	(0.03)		
Min(Mill Entrants,5)	0.14	(0.02)	0.09	(0.03)
Mill (Dummy) * (Mill Entrants=1)	-0.06	(0.07)		
Min(Logger Entrants,5)	0.06	(0.02)		
Potential Logger Entrants			0.01	(0.00)
Constant	2.71	(0.20)	2.20	(0.26)
<i>Poisson parameter and Weibull scale parameter include year dummies.</i>				
	<i>ln(p)</i>			
Mill(Dummy)	0.01	(0.07)		
Min(Mill Entrants,5)	0.06	(0.02)		
Mill (Dummy) * (Mill Entrants=1)	-0.08	(0.12)		
Min(Logger Entrants,5)	0.03	(0.02)		
Constant	0.94	(0.09)		
	<i>ln(θ)</i>			
Constant	-0.46	(0.13)		
	N=1325		N = 300	
	Wald χ^2 (23)	851.8	LR χ^2 (21)	199.5
	P-value	0.000	P-value	0.000
			Pseudo-R2	0.14

Note: Bid distribution estimated on sealed bid auctions with two or more bidders (255 auctions).

Figure 2A: Actual vs. Estimated Density of Sealed Bid Residuals (Northern Sales)

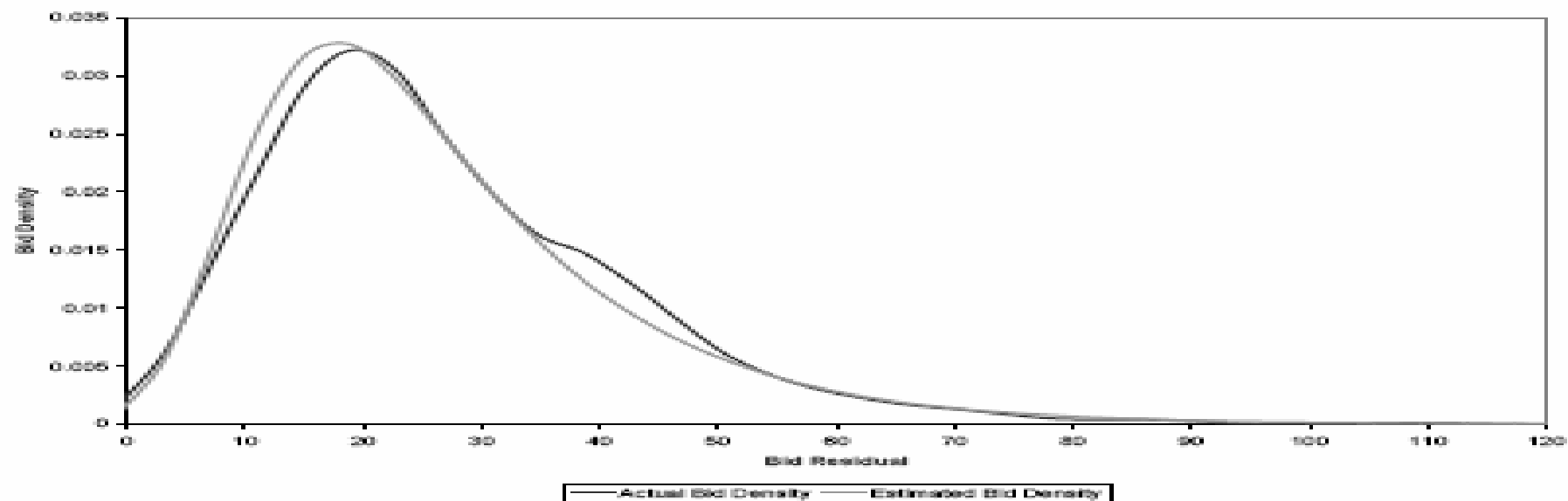
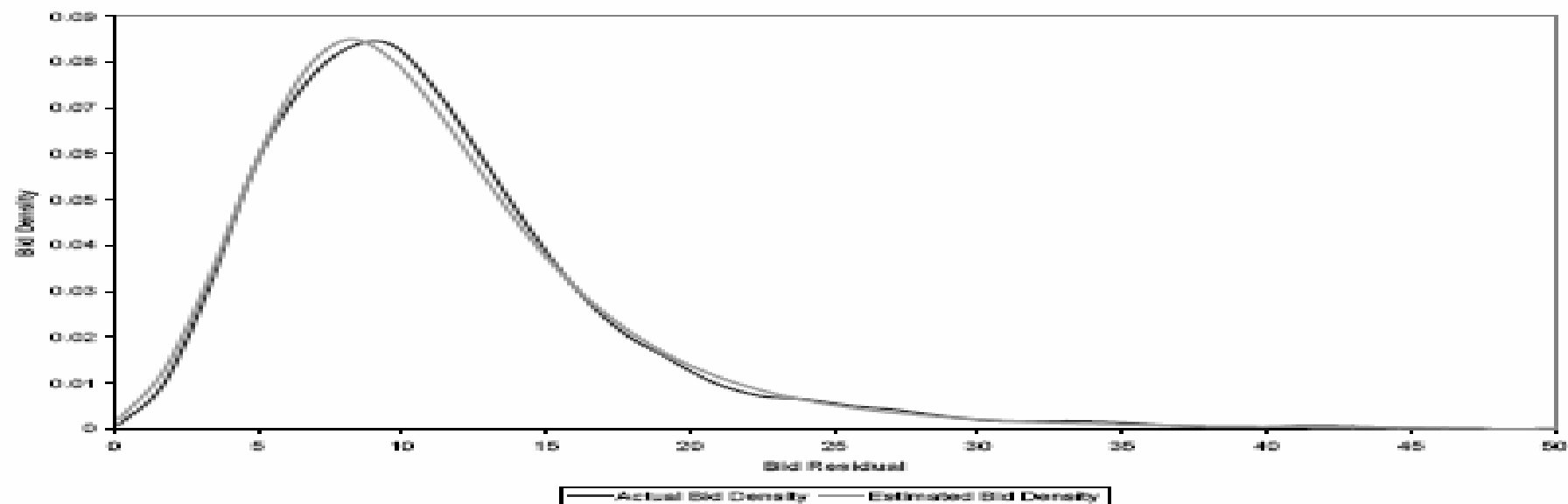


Figure 2B: Actual vs. Estimated Density of Sealed Bid Residuals (CA Sales)



Estimate Value Distributions

- Assumption:
observed bids are consistent with equilibrium behavior
- A Bidder's value v_i must be related to his observed bid b_i through FOC:

$$v_i = \phi_i(b_i; X, u, N, n) = b_i + \frac{1}{\sum_{j \in n \setminus i} \frac{g_j(b_i | X, u, N, n)}{G_j(b_i | X, u, N, n)}}.$$

Estimate Value Distributions (cont'd)

- We can recover value distribution F_L , F_M , as in Guerre, Perrigne, Vuong (2000)
- Use Krasnokutskaya (2002) for unobserved heterogeneity, u

$$F_k(v|X, u, N) = G_k(\phi_k^{-1}(v; X, u, N, n)|X, u, N, n).$$

Figure 3A: Estimated Value Distributions and Bid Functions for the Case of Two Loggers and Two Mills (Northern Sales)

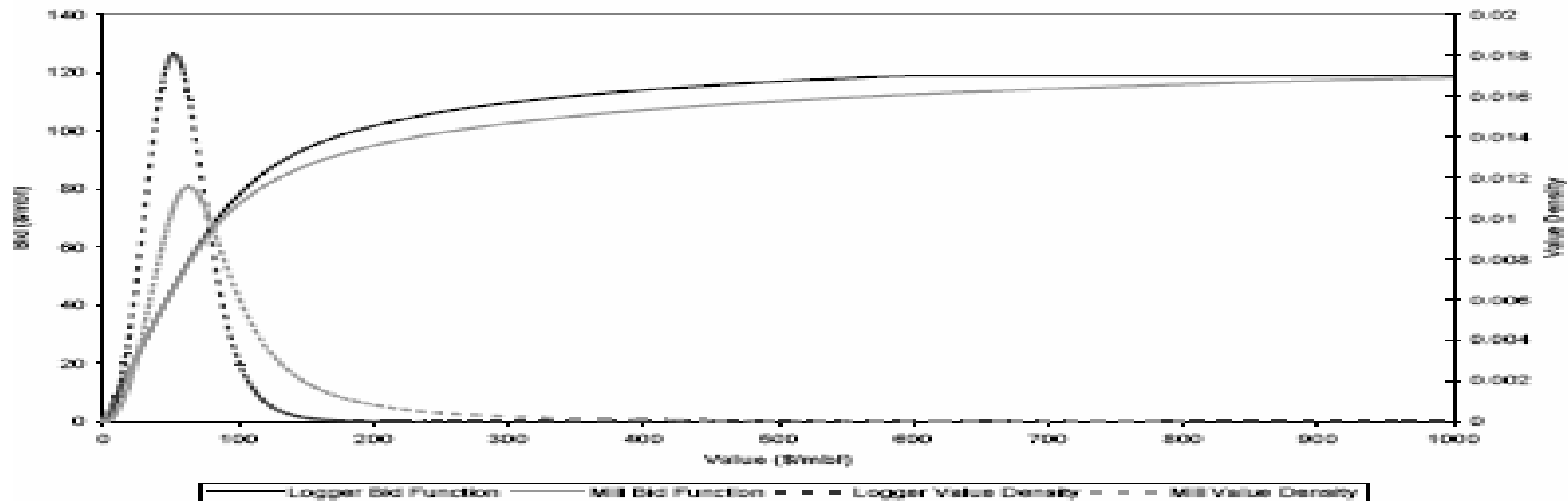
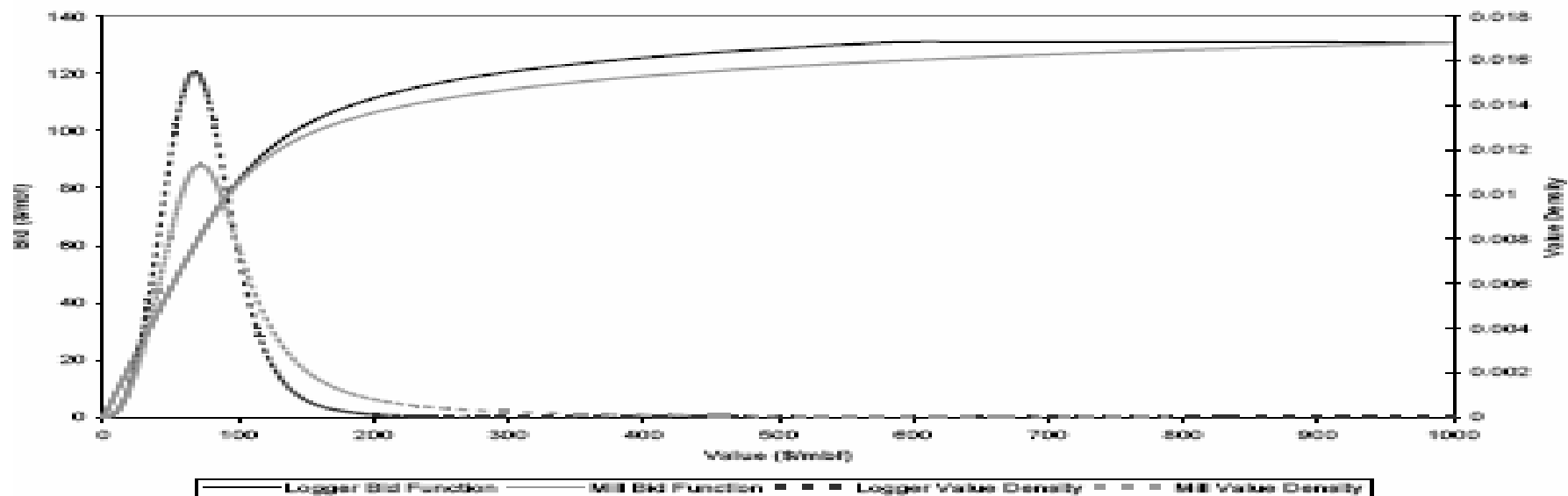


Figure 3B: Estimated Value Distributions and Bid Functions for the Case of Two Loggers and Two Mills (CA Sales)



Estimate Entry Costs

- Recover entry costs from equilibrium entry condition
- Average entry costs: $K(X, N)$
- This equals to

$$\Pi_L^s(X, N) = \sum_{n \subseteq N} \bar{\pi}_L^s(X, N, n) \Pr[n|X, i \in n, N, s] .$$

- Number of entering loggers: Poisson with mean

$$\mu(X, N) = X\alpha_X + N\alpha_X$$

Table 4A: Bid and Entry Distributions for Sealed Auctions (Northern Sales)

	(1)		(2)	
	Bid Distribution (Weibull)		Logger Entry (Poisson)	
	coefficient	s.e.	coefficient	s.e.
	<i>ln(λ)</i>		<i>ln(μ)</i>	
Ln(Reserve Price)	0.42	(0.04)	-0.29	(0.05)
Ln(Selling Value)	-0.01	(0.02)	-0.03	(0.03)
Ln(Manufacturing Costs)	0.44	(0.14)	0.85	(0.17)
Ln(Logging Costs)	-0.44	(0.14)	-0.81	(0.17)
Ln(Road Costs)	0.00	(0.02)	-0.16	(0.04)
Species Herfindal	-0.10	(0.11)	-0.24	(0.15)
Density of Timber (hmbf/acres)	-0.96	(0.31)	-0.93	(0.44)
Salvage Sale (Dummy)	-0.05	(0.05)	-0.02	(0.07)
Scale Sale (Dummy)	-0.05	(0.05)	-0.15	(0.08)
Ln(Volume)	-0.08	(0.03)	-0.24	(0.04)
Kootenai NF (Dummy)	0.14	(0.06)	0.18	(0.09)
Mill (Dummy)	0.27	(0.03)		
Min(Mill Entrants,5)	0.14	(0.02)	0.09	(0.03)
Mill (Dummy) * (Mill Entrants=1)	-0.06	(0.07)		
Min(Logger Entrants,5)	0.06	(0.02)		
Potential Logger Entrants			0.01	(0.00)
Constant	2.71	(0.20)	2.20	(0.26)
<i>Poisson parameter and Weibull scale parameter include year dummies.</i>				
	<i>ln(p)</i>			
Mill(Dummy)	0.01	(0.07)		
Min(Mill Entrants,5)	0.06	(0.02)		
Mill (Dummy) * (Mill Entrants=1)	-0.08	(0.12)		
Min(Logger Entrants,5)	0.03	(0.02)		
Constant	0.94	(0.09)		
	<i>ln(θ)</i>			
Constant	-0.46	(0.13)		
	N=1325		N = 300	
	Wald χ^2 (23)	851.8	LR χ^2 (21)	199.5
	P-value	0.000	P-value	0.000
			Pseudo-R2	0.14

Note: Bid distribution estimated on sealed bid auctions with two or more bidders (255 auctions).

Comparing Predicted and Actual Outcomes

- How closely calibrated model can explain open auction outcomes?
- How is the degree of bidder competitiveness?
- Sealed bidding predictions:
 - Estimated Poisson model: distribution of loggers
 - Use estimates of G_L and G_M , and distribution of u
 - Predict bidding behavior, compare

Comparing Predicted and Actual Outcomes (cont'd)

- Open auction
 - each entrant will bid his value
 - if collude, all mill but highest value drop out
- Predictions by simulations
 - draw u
 - draw value from either F_L or F_M
 - calculate price, profits, surplus

Table 5A: Actual Outcomes vs. Outcomes Predicted by Model (Northern Sales)

		(1)	(2)	(3)
	N	Actual	Predicted (bidding only)	Predicted (entry + bidding)
<i>Sealed Bid Sales</i>				
Avg. Bid	1370	68.5	67.9 (1.9)	67.2 (1.9)
Avg. Logger Bid	1027	57.6	56.6 (1.7)	55.8 (1.6)
Avg. Mill Bid	343	101.0	101.5 (4.2)	101.5 (4.2)
Avg. Sale Price	300	81.1	83.8 (2.5)	85.5 (2.6)
Avg. Revenue	300	116,207	112,392 (5,281)	116,053 (5,782)
% Sales won by Loggers	300	68.7	67.0 (1.2)	64.5 (1.3)
Avg. Logger Entry	300	3.42		3.42 (0.09)
<i>Open Auction Sales</i>				
Avg. Sale Price (Competition)	658	72.8	80.7 (2.5)	79.7 (3.6)
Avg. Sale Price (Collusion)	658	72.8	53.0 (1.5)	51.7 (2.6)
Avg. Revenue (Competition)	658	156,937	165,039 (8,758)	166,016 (9,375)
Avg. Revenue (Collusion)	658	156,937	63,507 (1,521)	66,627 (3,159)
% Sales won by Loggers	658	60.0	55.2 (1.2)	53.0 (2.4)
Avg. Logger Entry	658	2.84		2.89 (0.16)

Note: Bootstrap standard errors in parentheses.

Table 5B: Actual Outcomes vs. Outcomes Predicted by Model (California Sales)

		(1)	(2)	(3)
	N	Actual	Predicted (bidding only)	Predicted (entry + bidding)
<i>Sealed Bid Sales</i>				
Avg. Bid	1184	80.8	82.7 (4.1)	80.7 (3.6)
Avg. Logger Bid	812	68.2	68.1 (3.7)	65.2 (3.2)
Avg. Mill Bid	372	108.3	114.6 (5.7)	114.6 (5.7)
Avg. Sale Price	269	92.1	97.8 (4.0)	98.7 (4.7)
Avg. Revenue	269	107,354	116,682 (5,572)	118,887 (6,156)
% Sales won by Loggers	269	62.1	62.1 (1.7)	60.2 (1.8)
Avg. Logger Entry	269	3.02		3.02 (0.13)
<i>Open Auction Sales</i>				
Avg. Sale Price (Competition)	212	119.0	120.7 (5.7)	115.1 (8.7)
Avg. Sale Price (Collusion)	212	119.0	65.8 (2.8)	63.5 (4.0)
Avg. Revenue (Competition)	212	269,511	294,775 (18,923)	281,958 (23,285)
Avg. Revenue (Collusion)	212	269,511	111,860 (4,319)	123,397 (6,453)
% Sales won by Loggers	212	42.9	42.9 (1.8)	39.0 (4.7)
Avg. Logger Entry	212	2.12		1.87 (0.26)

Note: Bootstrap standard errors in parentheses.

Table 6A: Welfare Effects of Sealed vs. Open Auctions (Northern Sales)

	(1)	(2)	(3)		(4)		(5)	
	Sealed Bid	Open Auction (Competitive)	Difference		Open Auction (Part. Collusion)		Difference	
Predict Bidding Only								
Avg. Sale Price	82.0	81.5	0.5	(0.1)	75.1	6.9	(0.4)	
Avg. Sale Revenue	149,045	148,394	651	(141)	128,613	22,432	(1,752)	
Avg. Sale Surplus	251,908	252,008	-100	(42.7)	252,008	-100	(42.7)	
% Sales Won by Loggers	59.4	58.7	0.7	(0.2)	58.7	0.7	(0.2)	
Predict Entry & Bidding								
Avg. Sale Price	83.5	80.9	2.6	(3.1)	74.2	9.3	(3.0)	
Avg. Sale Revenue	154,302	148,959	5,343	(5,582)	127,272	27,030	(5,582)	
Avg. Sale Surplus	256,382	253,580	2,803	(9,900)	253,580	2,803	(9,900)	
% Sales Won by Loggers	58.7	56.0	2.6	(2.6)	56.0	2.6	(2.6)	
Logger Entry	3.28	2.98	0.30	(0.23)	2.98	0.30	(0.28)	

Note: Each entry is an average prediction over all tracts in the sample. Bootstrap standard errors in parentheses.

Table 6B: Welfare Effects of Sealed vs. Open Auctions (California)

	(1) Sealed Bid	(2) Open Auction (Competitive)	(3) Difference		(4) Open Auction (Part. Collusion)	(5) Difference	
			est.	s.e.		est.	s.e.
Predict Bidding Only							
Avg. Sale Price	108.2	107.5	0.6	(0.1)	97.2	11.0	(0.8)
Avg. Sale Revenue	195,833	194,815	1,018	(119)	166,964	28,869	(2,299)
Avg. Sale Surplus	289,776	289,821	-45	(23.3)	289,821	-45	(23.3)
% Sales Won by Loggers	53.9	53.3	0.5	(0.2)	53.3	0.5	(0.2)
Predict Entry & Bidding							
Avg. Sale Price	112.7	101.4	11.3	(6.7)	91.6	21.1	(6.2)
Avg. Sale Revenue	210,269	184,026	26,243	(12,821)	159,796	50,474	(11,872)
Avg. Sale Surplus	304,145	274,116	30,029	(52,426)	274,116	30,029	(52,426)
% Sales Won by Loggers	53.2	48.9	4.3	(4.5)	48.9	4.3	(4.5)
Logger Entry	2.84	2.26	0.58	(0.3)	2.26	0.58	(0.3)

Note: Each entry is an average prediction over all tracts in the sample. Bootstrap standard errors in parentheses.

Discussion

- “Independent value” assumption: Does it apply?
- Distributional assumptions: Strong?
- “Randomized” sample?